

MODEL ANALYSIS REPORT

Dhaka House Rent Prediction | Updated technical review

1. Executive Summary

The project estimates monthly apartment rent in Dhaka using grouped area, size, bedrooms and bathrooms. The raw dataset has 28,800 rows; after removing 13,541 duplicates and applying the notebook's filters, 15,001 rows remain across 69 grouped areas. The notebook compares four regressors and uses a train–test R^2 gap below 5 percentage points plus lowest MAE to select a model.

2. Dataset and Preprocessing

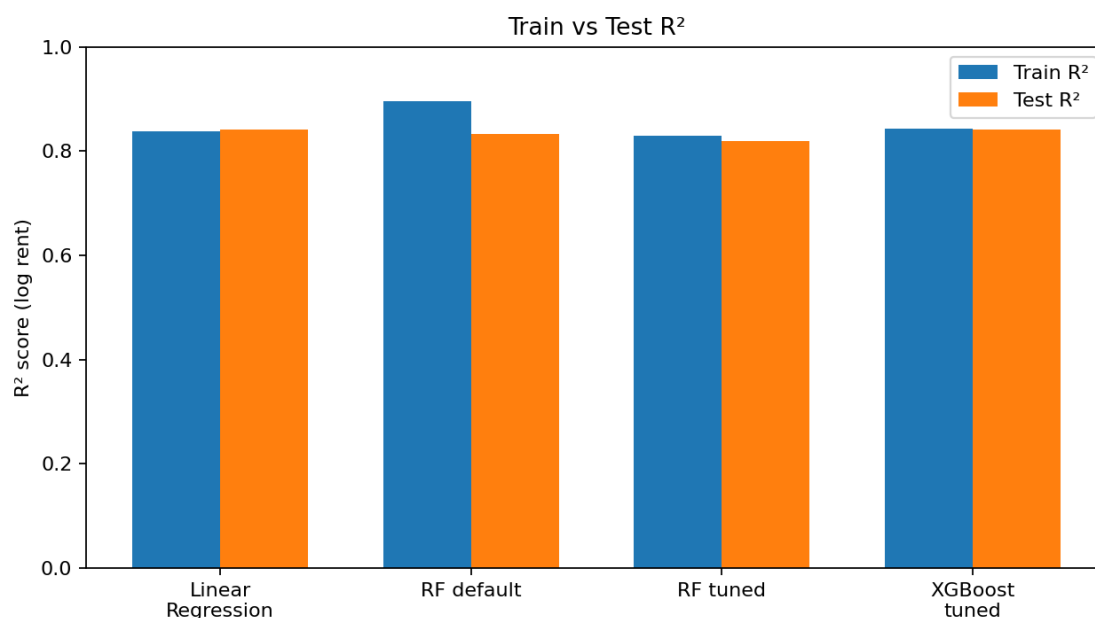
Item	Finding
Raw records	28,800
Duplicate rows removed	13,541
Rows after duplicate removal and filters	15,001
Grouped areas	69
Missing values in raw dataset	0
Filtered rent range	BDT 9,000 – 120,000
Median rent	BDT 17,000
Median apartment size	1,000 sqft

Cleaning converts rent and area strings to numbers, groups detailed locations into main areas, removes duplicate rows, filters rent outside the 1st–99th percentiles, and restricts size to 200–4,000 sqft. Statistics describe the filtered modelling dataset, not all current listings.

3. Model Evaluation

R^2 is measured on log-transformed rent. MAE and RMSE are calculated in BDT after inverse transformation.

Model	Train R^2	Test R^2	Gap pp	MAE BDT	RMSE BDT
Linear Regression	0.838	0.841	-0.3	3,387	6,483
Random Forest (default)	0.896	0.833	6.2	3,442	6,578
Random Forest (tuned)	0.829	0.819	1.0	3,572	6,609
XGBoost (tuned)	0.842	0.841	0.1	3,367	6,271



4. Interpretation

The notebook selection rule chooses XGBoost (tuned) in this reproduced run: Test $R^2=0.841$, MAE BDT 3,367, RMSE BDT 6,271, gap 0.1 percentage points. The default Random Forest's gap exceeds the notebook's 5-point threshold. These results apply to this random split and are not a guarantee of future performance.

5. Saved Model and App

The saved model is XGBRegressor, with 72 feature columns and 69 area labels. The Flask app predicts log rent, applies expm1 and rounds to BDT 100. Validations enforce size 200–4,000 sqft, bedrooms 1–6, bathrooms 1–7 and a known area.

6. Limitations and Recommendations

- Listings may not reflect final agreed rents; MAE is an average, not a per-property guarantee.
- Missing features include building age, floor, lift, parking, furnishing, condition, utilities and exact street.
- Use newer independent data and consider time-based or area-based validation.
- Report error by neighbourhood and rent band; add reliable property features when available.
- GitHub-ready app disables Flask debug mode. Load only trusted joblib files.
- Before making the repository public, verify rights to redistribute the dataset and model, and add the original data source/license.

7. Conclusion

This is a useful prototype for indicative monthly rent estimates. Validate on newer, independent rental data before relying on predictions for real-world decisions.